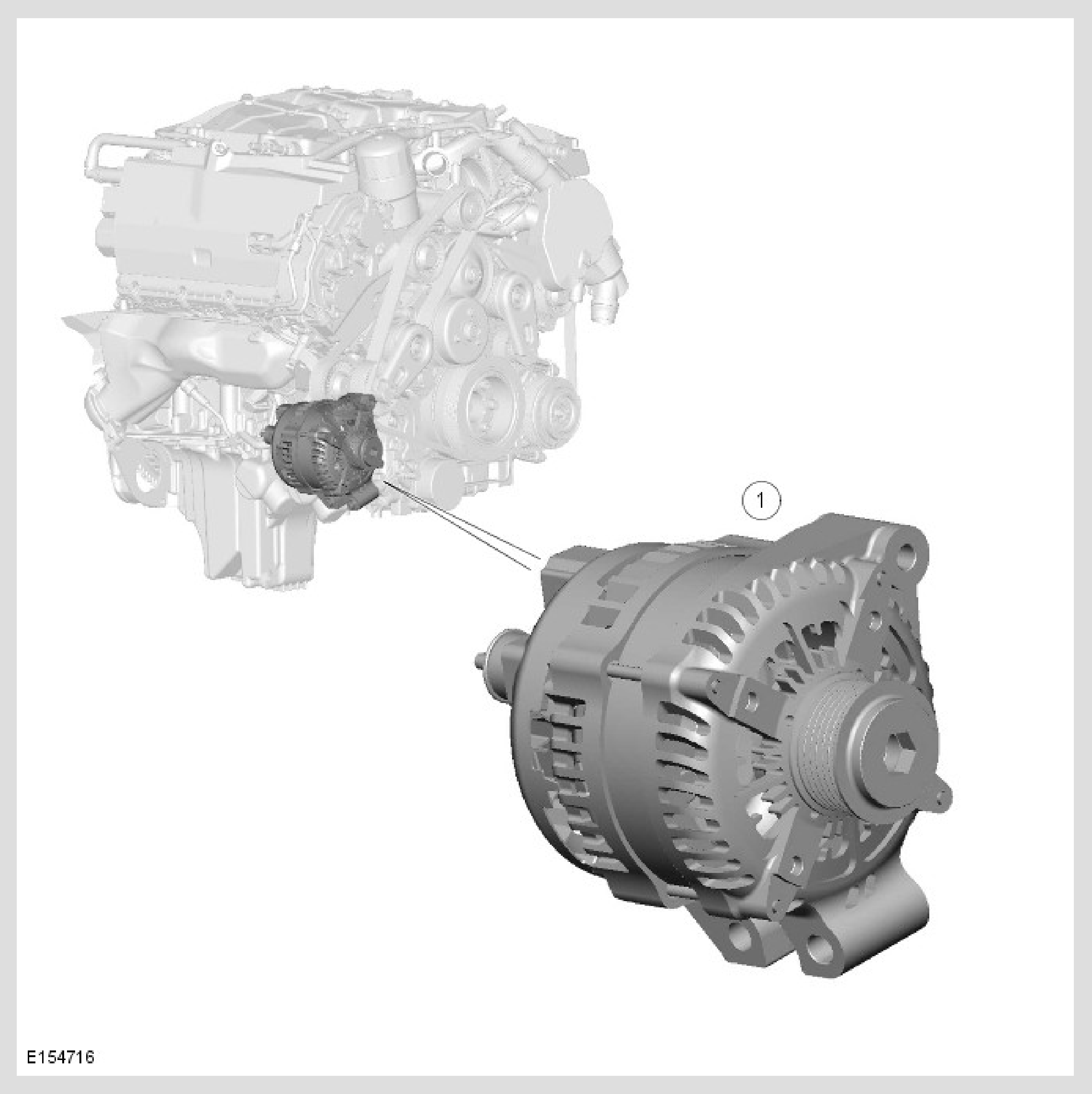
PUBLISHED: 21-AUG-2013
2014.0 RANGE ROVER (LG), 414-02

GENERATOR AND REGULATOR - V8 N/A 5.0L PETROL/V8 S/C 5.0L PETROL

COMPONENT LOCATION

NOTE:

V6 3.0L SC Petrol engine shown, V8 5.0L SC Petrol engine very similar.



OVERVIEW

The generator is mounted on the front right side of the engine and is driven at approximately three times engine speed by the accessory drive belt.

The charging system consists of a generator and regulator assembly and a GWM (gateway module). The generator and regulator assembly generates electrical power for the vehicle electrical system and maintains the primary battery and the secondary or auxiliary battery (where fitted) in a charged state. The rate of charge for the primary battery and the secondary or auxiliary battery (where fitted) is controlled by the GWM. For additional information, refer to: Battery and Cables (414-01, Description and Operation).

When the engine is running the generator produces an AC (alternating current) , which is converted to a DC (direct current) internally. The output from the generator is controlled by the voltage regulator (located inside the generator) and then supplied to the primary battery through the BJB2 (battery junction box 2) to the primary battery positive cable.

Charging of the secondary or auxiliary battery on dual battery vehicles is controlled by the GWM and the PSDB (power supply distribution box). Generator output is directed to the secondary or auxiliary battery via a contactor in the PSDB which is controlled by the GWM.

DESCRIPTION

The regulator provides a controlled variable DC voltage output from the generator. Two electrical terminals are provided on the outer casing of the generator. One terminal supplies the DC voltage output from the generator to the battery positive terminal. The second terminal provides the LIN (local interconnect network) bus connection between the regulator and the GWM (gateway module).

The output from the generator is AC . A bridge rectifier within the generator converts the AC output to DC.

The generator comprises a rotor winding, which is rotated by the generator pulley via the primary drive belt, a stator which is retained stationary within the generator housing, a voltage output regulator and a bridge rectifier.

The rotor winding assembly rotates inside the stator winding. The rotor generates a magnetic field and the stator winding develops AC voltage and current flows from the induced magnetic field of the rotor.

The regulator monitors the stator output voltages. Depending on the measured voltages, and signals from the GWM on the LIN bus, the regulator will adjust the amount of rotor field current to control generator output. The regulator is located within the generator housing.

The regulator will vary the charging system voltage output level depending on signals from the GWM. If the charging system voltage falls below a predetermined level, the GWM, via the LIN bus, will control the regulator to increase the field current produced by the stator and the rotor, increasing the magnetic field, which results in an increased generator output. If the charging system voltage raises above the predetermined level, the GWM, via the LIN bus, will control the regulator to decrease field current, reducing the magnetic field and decreasing the generator output. The regulator controls the field current by adjusting the power supplied to the rotor windings.

The following variants of generator are fitted to the specific engine types:

- TDV6 3.0L Diesel - 220A SC4 Denso
- TDV8 4.4L Diesel - 220A SC4 Denso
- V6 3.0L SC Petrol - 180A SC3 Denso
- V8 5.0L SC Petrol - 180A SC3 Denso

OPERATION

The output voltage required from the generator and regulator is calculated by the GWM (gateway module).
For additional information, refer to: Battery and Cables (414-01, Description and Operation).

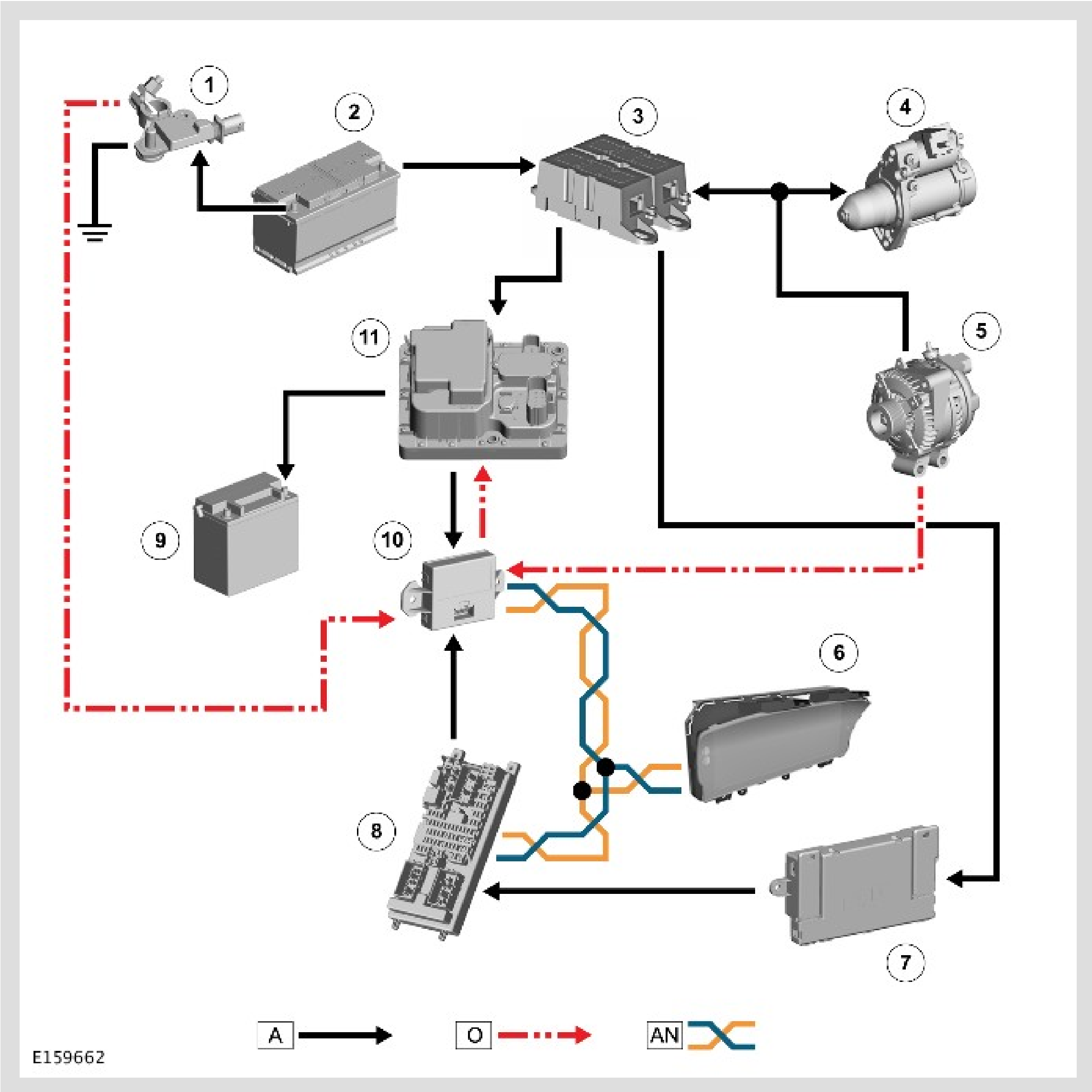
The BMS (battery monitoring system) control module on the primary battery, and the BMS control module on the auxiliary battery (if fitted), is connected to the GWM via a LIN bus connection. The BMS control module measures battery current and voltage, which it communicates to the GWM over a LIN bus connection. Based on the information received from the BMS control module, the GWM will control the output from the generator, and if necessary, will also request the switching off of electrical loads if necessary. The GWM controls the generator voltage regulator via the LIN bus connection and can adjust the generator output applicable to demand.

The GWM will over-ride the voltage value if it detects a fault in the power supply system.

The GWM also signals the instrument cluster to display a warning message if it detects a fault with the generator and regulator.
For additional information, refer to: Instrument Cluster (413-01, Description and Operation).

The GWM controls the charge warning indicator in the instrument cluster. Failure of the generator or regulator to provide sufficient charge for the primary battery to support vehicle demand, the GWM will instruct the instrument cluster via the HS (high speed) CAN (controller area network) powertrain bus to illuminate the charge warning indicator.

CONTROL DIAGRAM



A = HARDWIRED; O = LIN BUS; AN = HS (HIGH SPEED) CAN POWERTRAIN BUS.

ITEM	DESCRIPTION
1	Battery Monitoring System (BMS) control module - primary battery
2	Primary battery
3	Battery Junction Box 2 (BJB2)
4	Starter motor
5	Generator
6	Instrument Cluster (IC)
7	Auxiliary Junction Box (AJB)
8	Central Junction Box (CJB)
9	Secondary or auxiliary battery

10	Gateway Module (GWM)
11	Power Supply Distribution Box (PSDB)